SIH 2026 · PROBLEM STATEMENT 168 · MODEL REPORT

Smartphone dead reckoning through GNSS outages

A residual Temporal Fusion Transformer cleans the phone's IMU. A 9-state EKF integrates it, and an OSM particle filter keeps it on real roads. This final report covers the architecture, the method, how the evaluation was made leak-free, where the remaining drift comes from, and what was tried.

Dataset IO-VNBD · 10 Hz · 60 s outages · median drift = end-point error ÷ distance travelled · final results, 4 Oct 2026 · protocol: causal pre-outage calibration + OSM planned route (no GNSS after outage start)

12.4%

Median 60 s drift, held-out test drives Y1 + M

3 seeds, range 11.7–12.9 · target < 10% not met

7.8%

Median 60 s drift, held-out drive Vw4

3 seeds, range 7.6–7.9 · never used for tuning

9.3%

Median 60 s drift, validation drives S2 + Vtb5

3 seeds, range 9.0–9.5 · used for model selection

492k

Model parameters

3.8 MB ONNX · CPU · edge-deployable

1. The problem

When a phone loses GNSS (tunnels, urban canyons, multi-storey car parks), the app has to keep estimating position from the phone's own accelerometer and gyroscope. Integrating raw phone IMU twice drifts fast. On our held-out test windows it reaches a median of **28.5%** of the distance travelled after 60 s (measured under the earlier evaluation protocol). The problem statement asks for drift below 10%, map matching to real roads, and a model small enough to run on the phone.

We use the IO-VNBD dataset: phone (S) and vehicle (V) CSV logs at 10 Hz with GNSS ground truth. Two data problems had to be fixed before any learning:

- **Clock offsets.** Phone and vehicle clocks disagreed by 0.3–116 s. We re-synchronised each phone log by cross-correlating its gyro yaw rate against GNSS-derived yaw rate.

- **Unknown phone mounting.** The phone's axes aren't aligned with the car. Before each outage we estimate the rotation, gyro bias and accelerometer bias by regressing the phone IMU against GNSS-derived vehicle motion, using **only data recorded before the outage starts** (section 6b).

Drives are split three ways. Training uses most drives. Validation uses **S2 + Vtb5** only: two other validation drives were in the training set of the current checkpoint, so they're excluded from model selection. The test drives **Y1, M and Vw4** are touched only for final reporting.

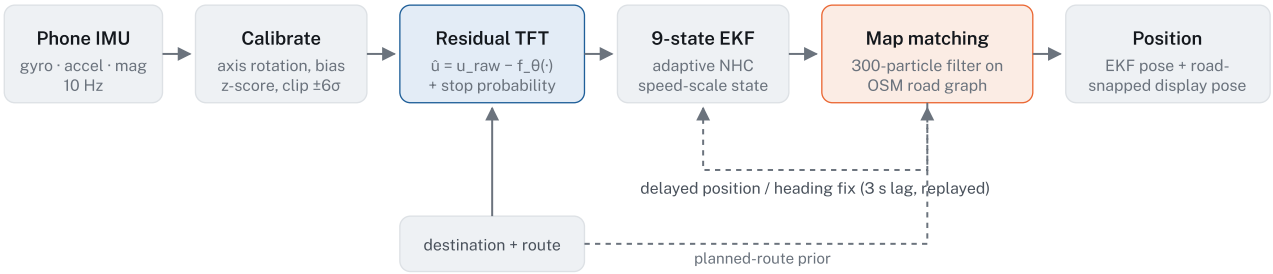
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2. End-to-end pipeline

Each stage has one job. The network never outputs a position; it outputs corrected sensor readings. Physics and the filters turn those into a trajectory, which keeps the output physically plausible and lets the map correct it.

Data flow during a GNSS outage

Solid arrows run at 10 Hz. Dashed arrows are 1 Hz map feedback applied with a 3 s delay and replayed forward.



The TFT, EKF and map matcher each see the planned route: the network as route geometry features, the particle filter as a prior bonus on route edges. All three are optional, so the pipeline degrades gracefully when no destination is set.

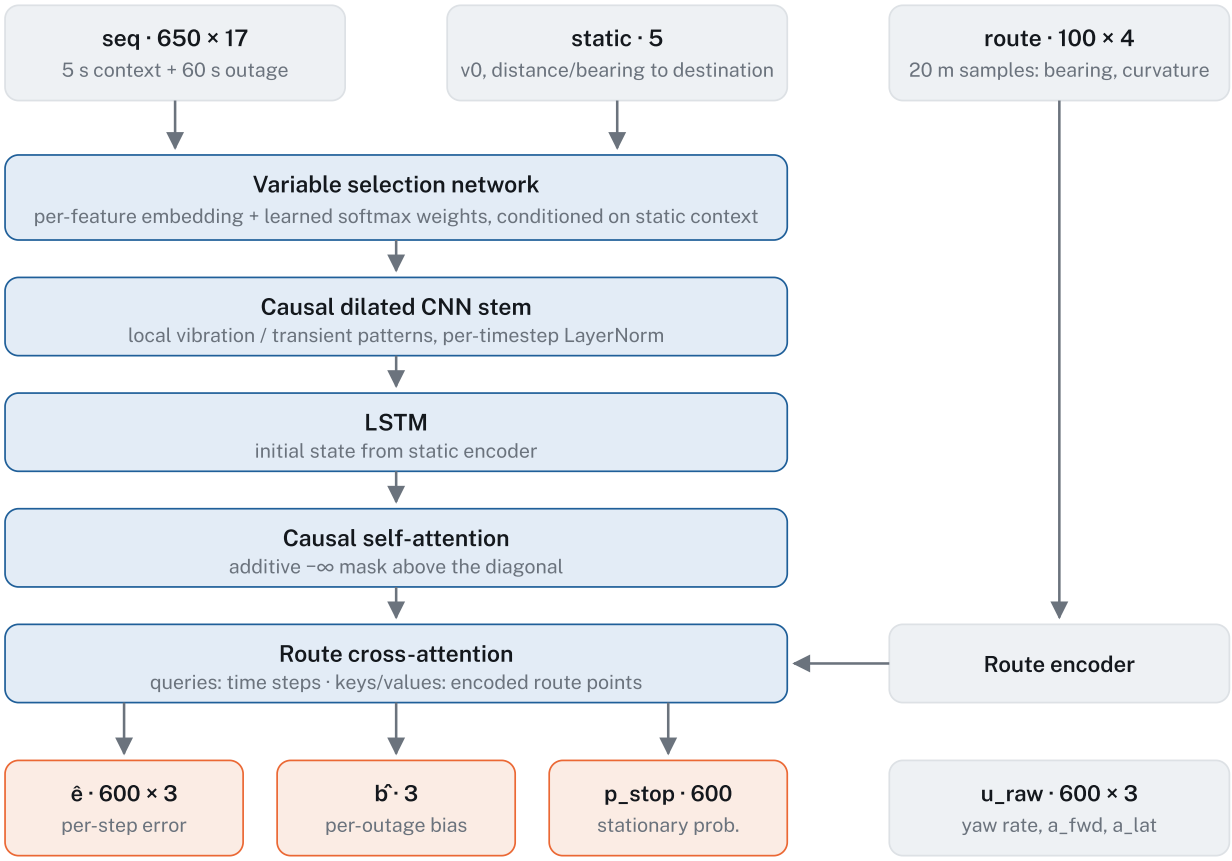
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3. Model architecture

The network is a **residual** model. It predicts the phone's sensor error, which is then subtracted from the raw reading. It doesn't predict velocity or position directly. A model that only has to learn corrections starts from a sensible answer (the raw IMU) and generalises better than one that has to learn motion from scratch.

ResidualTFT, non-autoregressive, whole 60 s outage at once

Input shapes for one sample. Every temporal layer is causal: step k never sees data after k.



`û = u_raw - (ê + b̂) → physics integration`

Why non-autoregressive: the whole outage is predicted in one forward pass, so errors in early predictions are never fed back as inputs. That removes exposure bias and the need for scheduled sampling. **Why causal:** at run time the phone only has data up to "now", so the model is trained under the same constraint, and future steps are zero-padded.

Physics integration (same equations in training and in the app)

```
Δψ_k = -Σ_{j≤k} ω̂_j · Δt
v_k  = relu( v0 + Σ_{j≤k} â_fwd,j · Δt ) · (1 - p_stop,k)
p_k  = p0 + Σ v_k · [sin ψ_k, cos ψ_k] · Δt          Δt = 0.1 s
```

Because integration is differentiable, the losses are computed on the resulting trajectory as well as on the corrected sensor values.

Training objective

- **Velocity-direction loss:** minimises the angle between predicted and true velocity vectors, penalising heading drift where it originates.
- **Increment losses** on per-step heading change and speed change, weighted by sample (`w_speed`) so braking and turning segments aren't drowned out by cruising.
- **Speed magnitude** and **stop** losses (`p_stop` against zero-speed labels).
- **Residual regularisation:** keeps ê small and smooth, so the model corrects the sensor rather than inventing motion.

Each training window gets a **random destination** taken from later in the same drive, with the route between them, so the model learns to use a destination when one exists and to cope without one. On test windows, route input lowers open-loop drift from 18.9% (destination only) to 18.1%.

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4. Filtering and map matching

The TFT output is integrated by a 9-state planar EKF. It carries what the network can't know from a single window: the slowly varying biases and the accumulated uncertainty.

STATE	MEANING	ROLE
`E, N`	Position (east, north)	What the app shows
`ψ`	Heading	Integrated from corrected yaw rate
`v_f, v_l`	Forward and lateral speed	v_l ≈ 0 enforced by the non-holonomic constraint (NHC)
`b_g, b_a`	Gyro and accel bias	Made observable by road-heading updates
`ds, b_s`	Speed scale and offset	Absorbs systematic speed error

Adaptive NHC. A car doesn't slide sideways, so lateral speed is pseudo-measured as zero. The noise on this constraint grows with yaw rate, acceleration and vibration, with Student-t reweighting of outliers, so it's trusted on straight roads and relaxed in hard turns.

Map matching. 300 particles live on the OSM road graph extracted per drive. Each particle has its own road, arc position, lateral offset and speed-scale error. Once a second each is weighted by:

- cross-track distance to its road;
- along-track agreement with the EKF;
- heading against the road bearing;
- curvature against the gyro;

- a large penalty for travelling against a road's direction.

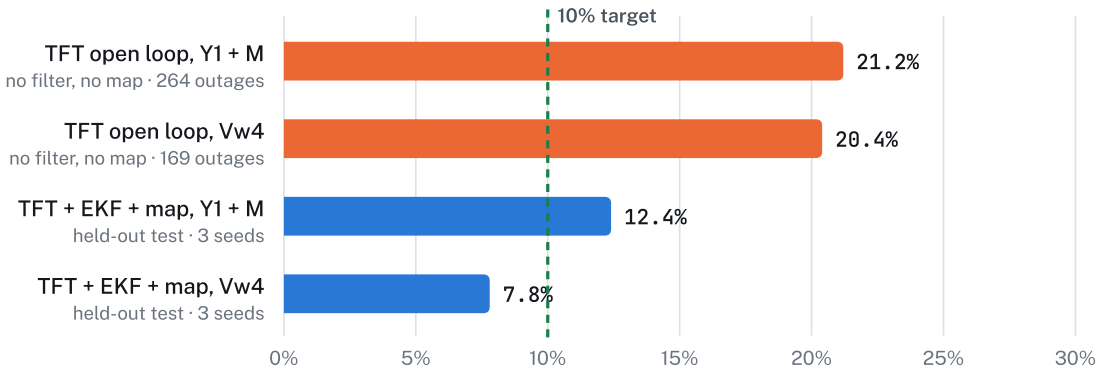
At junctions particles branch in proportion to heading agreement, with a planned-route bonus. The filter decides with a **3 s fixed lag**. When confident, it sends a position (and on straight roads, heading) fix back to the EKF, applied at the lagged time and replayed to the present.

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5. Drift results

Median 60 s drift at each stage

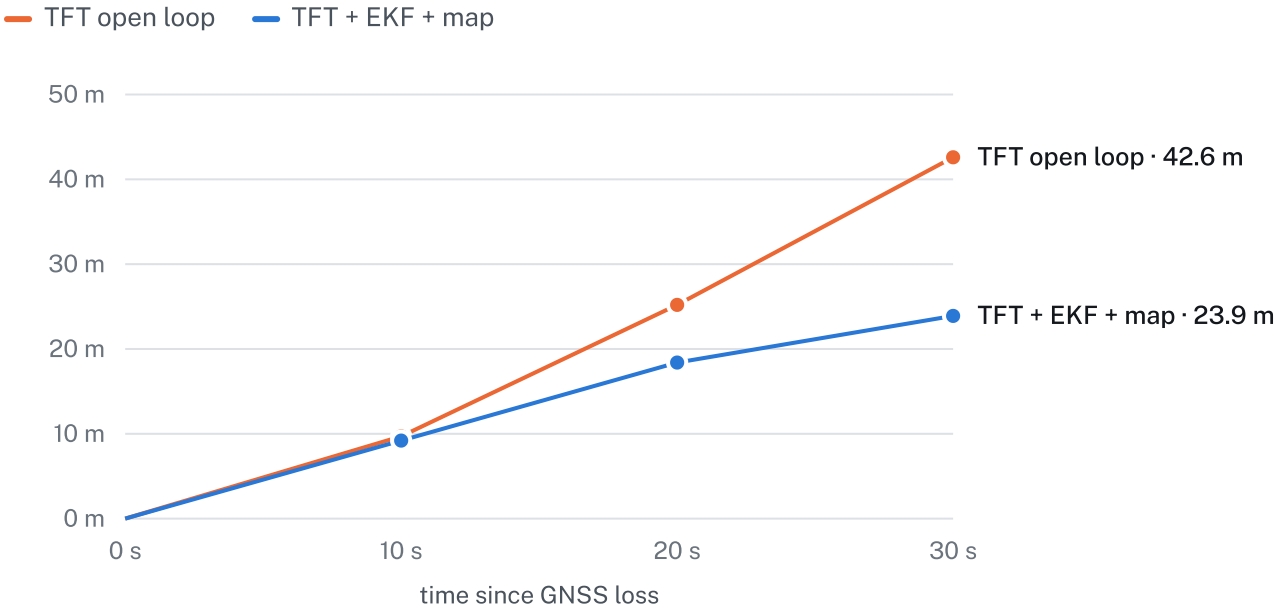
Lower is better. Target: below 10% of distance travelled.



Final leak-free protocol. Outages whose ground truth stays within 15 m of a mapped road, excluding cold-start outages with no usable pre-outage calibration (264 for Y1 + M, 169 for Vw4). Open-loop rows are the TFT-corrected IMU integrated without filter or map on the same outages; map-aided rows are the mean of 3 particle-filter seeds.

How error grows during an outage, Y1 + M

Median position error in metres after 10, 20 and 30 s.



Without the map, error grows almost linearly with time: a speed error integrates into a distance error. Map matching removes most cross-track error and bends the curve after the first junction.

SPLIT (60 S OUTAGES)	MEDIAN DRIFT	SEED RANGE	P90	< 10% DRIFT	WRONG ROAD	ALONG / CROSS
Test Y1 + M (264 outages)	12.4%	11.7–12.9	58%	45%	20.1%	53 / 8 m
Test Vw4 (169 outages)	7.8%	7.6–7.9	44%	58%	14.0%	67 / 6 m
Validation S2 + Vtb5 (223 outages)	9.3%	9.0–9.5	47%	53%	14.6%	49 / 5 m

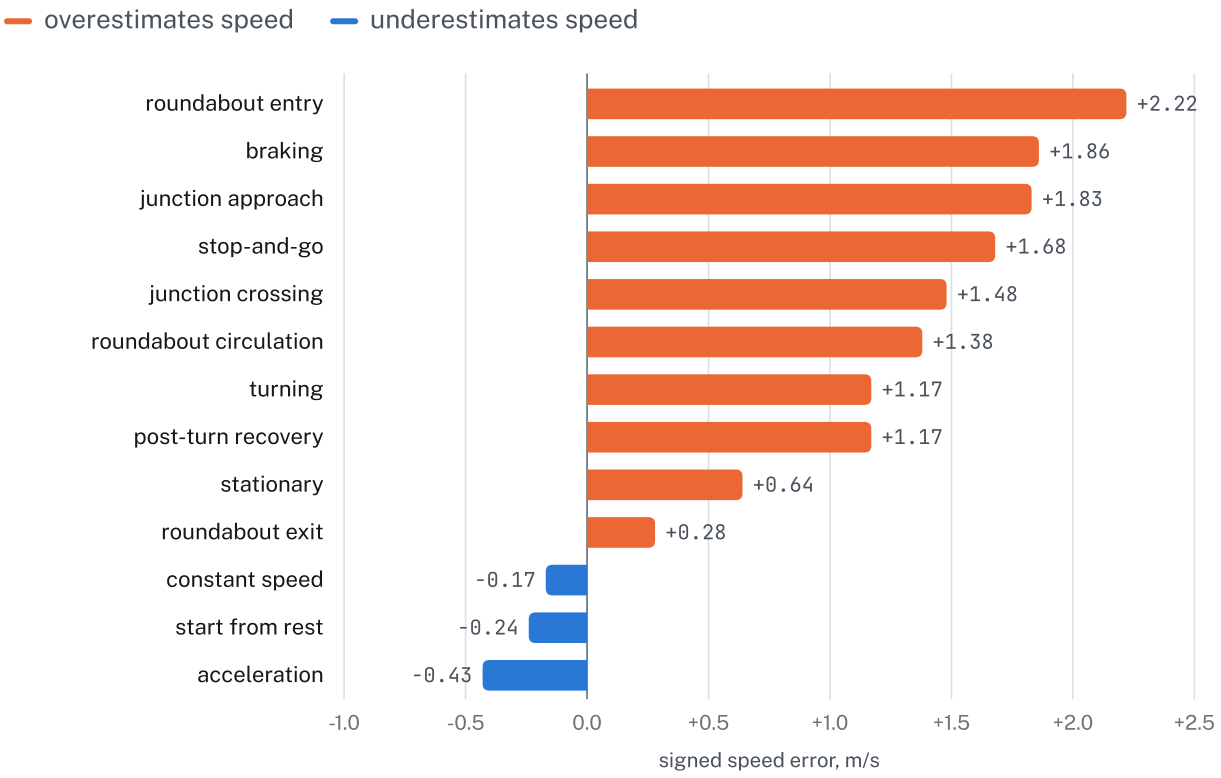
Under the leak-free protocol the 10% median target is met on the validation drives and on Vw4, but **not on Y1 + M (12.4%)**. With the earlier, leaky protocol Y1 + M measured 9.6%: whole-recording calibration had been hiding a real loss there (open-loop TFT drift 17.3% → 21.2%). Across all drives only about half of the outages are individually under 10%, and the worst tenth drift 44–58%. The rest of this report is about that tail.

6. Where the remaining drift comes from

Once the map fixes cross-track error, almost all remaining error is **along-track**: the car is on the right road but the estimate is ahead of or behind it. The median along-track error is about 45 m, against 5–7 m cross-track. Along-track error is speed error integrated over time, so we sliced the TFT's speed error by driving condition on the validation drives.

TFT speed bias by motion phase

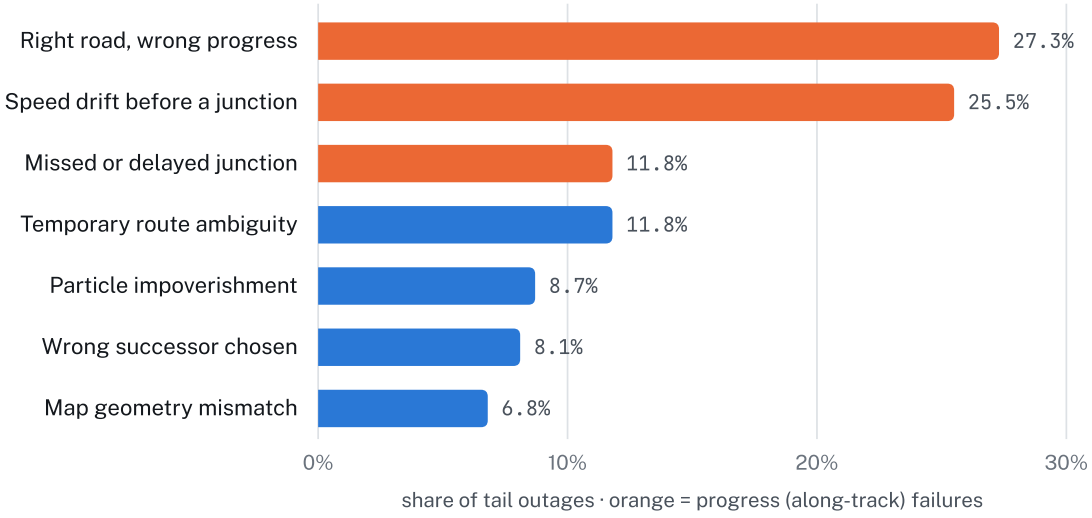
Mean signed speed error (m/s), validation drives. Positive means the model thinks the car is faster than it is.



Diagnosis (open-loop TFT, measured before the evaluation fixes; the same pattern holds afterwards, with map-aided braking at +1.0 m/s and roundabout entry at +1.1 m/s): the model is accurate at constant speed (−0.17 m/s) but responds too slowly to deceleration. It keeps too much speed when braking, approaching junctions and entering roundabouts, lagging the truth by 0.2–1.5 s. Across speeds it regresses to a typical value: +2.0 m/s at 3–8 m/s, −0.5 m/s above 15 m/s. Vibration, jerk and timestamp quality made little difference.

What goes wrong in the worst quarter of outages

Validation S2 + Vtb5: 161 tail outages, each assigned one failure class.



About two thirds of the tail is a progress error: the right road but the wrong place along it, often compounded at the next junction. In 74% of tail outages along-track error appears first, a median 15 s before any wrong turn. Wrong turns are mostly a *consequence* of speed drift, not an independent failure.

6b. Making the evaluation leak-free

An "IMU-only" result is only meaningful if nothing measured after GNSS is lost reaches the pipeline. An audit found two places where it did, plus one routing bug. All numbers in this report are from after the fixes.

ISSUE	FIX	EFFECT ON VALIDATION (S2 + VTB5, MAP-AIDED)
Phone calibration (axes, gyro bias, accelerometer bias) was fitted on whole recordings, including GNSS from inside each outage	Per-outage calibration from data before outage start minus 0.5 s only; outages without enough history are marked cold-start and reported separately (4 of 227)	9.6% → 9.2%: within seed noise. On test drives Y1 + M the two fixes together moved 9.6% → 12.4%
The TFT's route input was the next 2 km of the *actual* driven track	Shortest OSM route from the last fix to the user's destination; never compared with the true path. It matches the driven path in 72% of outages	9.2% → 9.3%; open-loop TFT 16.6% → 17.5%

ISSUE	FIX	EFFECT ON VALIDATION (S2 + VTB5, MAP-AIDED)
The map matcher's planned route silently fell back to "start road only" when the destination road was unreachable	Same planner as the TFT route; returns no route on failure	headline variant unaffected

The fixes matter most where the phone mount or sensor bias changes during a recording, which a whole-recording fit averages away. Leakage regression tests now poison all GNSS after outage start and check that calibration, model inputs, model outputs and filter tracks are bit-identical. The old behaviour is only reachable through explicit diagnostic flags, and the comparison tools refuse to mix the two conditions.

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7. What we tried, and what we kept

Every change is an ablation against the same control on the validation drives (S2 + Vtb5), with 3 particle-filter seeds and paired comparisons per seed. A change is kept only if it improves the worst-quarter tail consistently without worsening wrong-road, cross-track or cruising-speed bias. The last experiments were also checked once on the held-out drive Vw4. Seed-to-seed noise is about ±0.3–1 point, so small gains are treated as no change.

CHANGE	EFFECT	DECISION
Heading-only map updates (config B)	8.6% on Y1 + M, but 10.9% on the unseen Vw4	rejected, overfit
Speed calibration (affine / isotonic)	small median gain, worse tail	rejected
Deceleration-weighted retrain	braking bias +1.9 → +0.5 m/s, but cruising speed too low; map drift 11.1%	rejected
Multitask roundabout model	worse map-aided drift	rejected
Turn landmark (particle turn records, then map-located turns)	wrong-road slightly lower, but worst-quarter tail worse	rejected
Confidence-gated turn landmark (A3-gate)	tail change −20, +9, +23 m across seeds: inconsistent	rejected

CHANGE	EFFECT	DECISION
Confidence policy that softens map updates (A3-full)	drift 12.9%, wrong-road 33.6%: the map must keep full strength	`rejected`
Map look-ahead slowdown as acceleration (L1 / L2)	L1 too weak; L2 fixes braking bias but over-slows the car (one outage lost 12.75 m/s)	`rejected`
Look-ahead as a bounded speed pseudo-measurement + post-event recovery	validation: tail 142 → 130 m in every seed, wrong-road 14.6 → 13.2%; **held-out Vw4: wrong-road 14.0 → 17.0%, worse in every seed**	`experimental, off`
Pre-outage speed-bias gate for the look-ahead	on validation, no threshold kept the tail gain; stopped	`unfinished, off`

Outcome: none of the post-baseline changes passed both the validation and the held-out checks, so the shipped configuration is the baseline TFT + EKF + full-strength map. The look-ahead prior is the most promising lead: it reliably shrinks the worst-quarter error but slows particles at branch points on drives whose speed already reads low. All experimental features remain in the code behind flags that are off by default.

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8. Deployment

- Exported to ONNX (opset 17, 3.8 MB), with a manifest listing input layout, feature order, normalisation statistics and filter configuration.
- **ONNX Runtime must use `ORT_ENABLE_BASIC`.** The higher optimisation levels contain an operator-fusion bug (onnxruntime 1.29) that corrupts this model's outputs by up to 0.6. With BASIC, outputs match PyTorch within 2.3×10^{-6}.
- The app asserts the optimisation level at startup and runs four stored real windows as a self-check. A CI test compares PyTorch and ONNX on random and real inputs.
- The EKF and particle filter are plain NumPy-level arithmetic (300 particles, 1 Hz map step) and port directly to the app.

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9. Limitations

- Map-aided figures cover outages on mapped roads. Off-map driving (car parks, private roads) falls back to TFT + EKF, which is about 17–20% open loop.

- The held-out set is three drives, and seed-to-seed and drive-to-drive variation (±0.3–1 point) is similar in size to many candidate improvements.
- The dataset's phone/vehicle clock re-sync uses whole recordings. It repairs a logging fault a real phone wouldn't have, but it also decides which outages are usable.
- Normalisation statistics differ slightly from those the checkpoint was trained with (two validation drives were included then). The effect is small and documented.
- The 10% median target is met on Vw4 (7.8%) and the validation drives (9.3%) but not on Y1 + M (12.4%), and the tail (p90 of 44–58%) is not yet good enough for turn-by-turn guidance through long outages. Speed error while slowing down remains the main cause.